

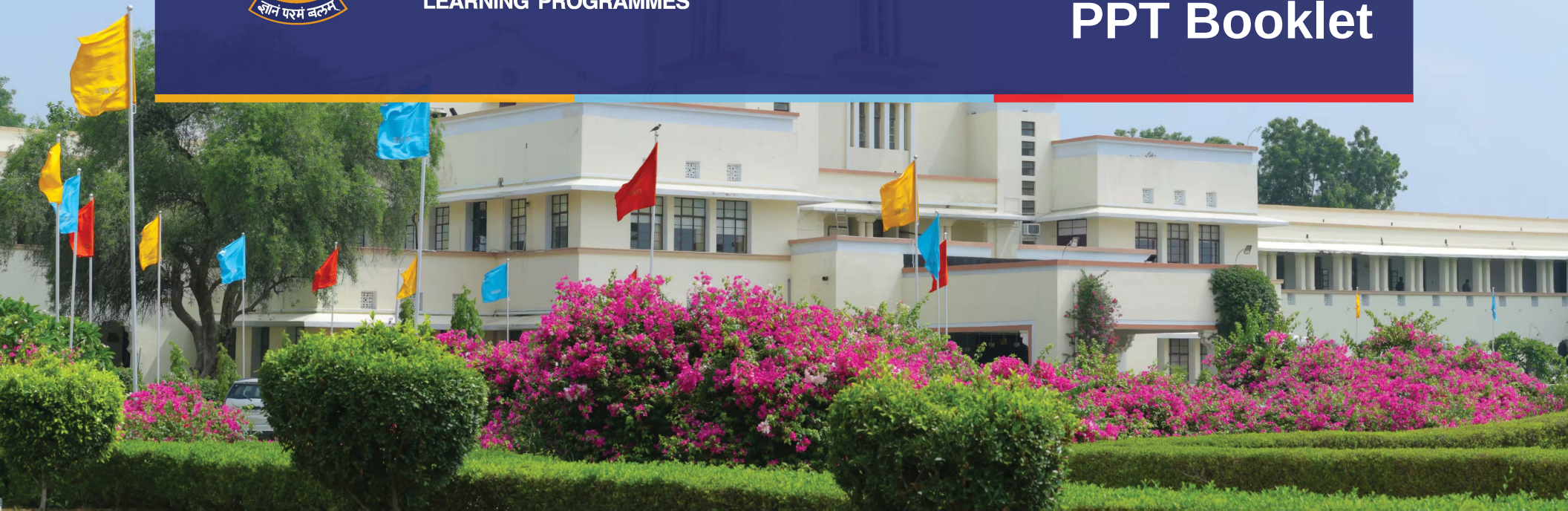


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**WORK INTEGRATED
LEARNING PROGRAMMES**

Wireless & Mobile Communication PPT Booklet





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IMP Note to Students

- It is important to know that just login to the session does not guarantee the attendance.
- Once you join the session, continue till the end to consider you as present in the class.
- IMPORTANTLY, you need to make the class more interactive by responding to Professors queries in the session.
- **Whenever Professor calls your number / name ,you need to respond, otherwise it will be considered as ABSENT**



**Wireless & Mobile Communication
(CSI ZG520)**

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Introduction

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Course Objective



- CO1** To learn the fundamentals of Mobile telecommunication systems and wireless networks.
- CO2** To learn different modulation schemes and radio propagation models used in communication systems.
- CO3** To learn multiple access techniques used in wireless networks.
- CO4** To learn modern wireless communication systems.
- CO5** To learn traffic routing in wireless networks.

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Learning Outcomes

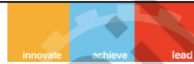


- LO1** Students will be able to acquire the concepts of wireless and mobile communication systems.
- LO2** Students will be able to differentiate various cellular technologies, modulation techniques and radio propagation models used in communication systems.
- LO3** Students will be able to describe multiple access techniques and equalization, diversity and channel coding techniques used in wireless communication.
- LO4** With the gained knowledge, students will be able to compare competitive technologies and the trade-offs associated.

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Modular Content Structure



M1: Introduction to Wireless Systems and Modern Wireless Communication System.

- M2: The Cellular Concept- System Design Fundamentals
- M3: Cellular Network
- M4: Mobile Radio Propagation: Large scale path Loss_ Part 1
- M5: Mobile Radio Propagation: Large scale path Loss_ Part 2
- M6: Mobile Radio Propagation: Large scale path Loss_ Part 3
- M7: Small Scale fading_ Part 1
- M8: Small Scale fading_ Part 1

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Modular Content Structure



- M9: Modulation Techniques for Mobile Radio_ Amplitude Modulation
- M10: Modulation Techniques for Mobile Radio_ Angle Modulation
- M11: Digital Modulation Techniques_1
- M12: Digital Modulation Techniques_2
- M13: Equalization & Diversity Techniques
- M14: Concepts of OFDM & MIMO
- M15: Multiple Access Techniques
- M16: Wireless Networking

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Text Book:

T1: “Wireless Communication Principles and Practice” by Theodore. S. Rappaport Second Ed., Prentice Hall of India, 2008.

T2: “Wireless Communications and Networks” by William Stallings – Pearson Education Ltd.

R1: “Mobile Communication” by Jochen H. Schiller, Addison –Wesley, Pearson Education Ltd., 2000.

R2: “Wireless and Mobile Network Architectures” by Yi-Bing Lin and Imrich Chaltamac, John Wiley and Sons, 2001.

M1: Introduction to Wireless Systems and Modern Wireless Communication System.

Type Description/Plan/Reference

RL 1.1 Introduction to wireless and Mobile communication

RL 1.2 Types of wireless communication

RL 1.3 A Modern wireless communication system

CS 1 A brief summary of recorded lectures, Examples of wireless communication system- paging systems, wireless local loop, WLAN, Bluetooth, Prioritizing handoff

HW 1 T1: Problems: 1.2, 1.5, 1.9, 1.16, 2.1, 2.2, 2.8

LE 1 Introduction to MATLAB/Simulink

Programming Language:

1. MATLAB/SIMULINK.
2. Python - NumPy Package.

Introduction

- Wireless communications has been associated with cellular telephony, as this is the biggest market segment, and has had the highest impact on everyday lives.
- In recent times, wireless computer networks have also led to a significant change in working habits and mobility of workers – answering emails in a coffee shop has become an everyday occurrence.
- But besides these widely publicized cases, a large number of less obvious applications have been developed, and are starting to change our lives.



- This variety of new applications causes the technical challenges for the wireless engineers to become bigger with each day.
- To give an overview of the solution methods for current as well as future challenges.
- There are two paths to developing new technical solutions:
- Engineering Driven and Market Driven.
- In the first case, the engineers come up with a brilliant scientific idea – without having an immediate application in mind.

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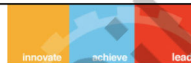
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- As time progresses, the market finds applications enabled by this idea.
- In the other approach, the market demands a specific product and the engineers try to develop a technical solution that fulfills this demand.

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- Market Demands.
- We start out with a brief history of wireless communications, in order to convey a feeling of how the science, as well as the market, has developed in the past 100 years.
- Then follows a description of the types of services that constitute the majority of the wireless market today.
- Each of these services makes specific demands in terms of data rate, range, number of users, energy consumption, mobility, and so on.
- A description of the interaction between the engineering of wireless devices and the behavioral changes induced by them in society.

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How It All Started

- Wireless communication is the oldest form of communication like shouts and drums did not require any wires or cables to function.
- Wireless communications started only with the work of Maxwell and Hertz, who laid the basis for the understanding of the transmission of electromagnetic waves.
- In 1898, Marconi, made his well-publicized demonstration of wireless communications from a boat to the Isle of Wight in the English Channel.
- It is noteworthy that while Tesla was the first to succeed in this important endeavor, Marconi had the better public relations, and is widely cited as the inventor of wireless communications, receiving a Nobel prize in 1909.

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Types of Services

- Broadcast
- Paging
- Cellular Telephony
- Trunking Radio
- Cordless Telephony
- Wireless Local Area Networks
- Personal Area Networks
- Fixed Wireless Access

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Broadcast

The first wireless service was broadcast radio.

Four properties differentiate broadcast radio

1. The information is sent unidirectional, i.e., in one direction only.
2. The information transmitted is same for all receivers.
3. The information is transmitted continuously.
4. In many cases like TV or radio, multiple transmitters send the same information

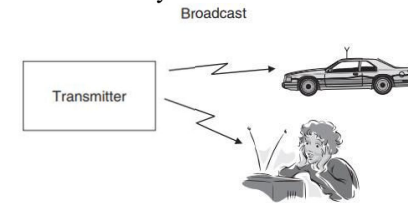


Figure Principle of broadcast transmission.

Paging

Similar to broadcast, paging systems are unidirectional wireless communications systems. The following properties characterize them:

1. The user can only receive information, but cannot transmit.
2. The information is intended for and received by, only a single user.
3. The amount of transmitted information is very small.

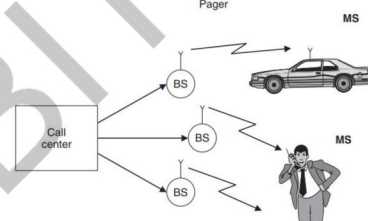


Figure Principle of a pager.

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Cellular Telephony

Cellular telephony is the economically most important form of wireless communications. The information flow is bidirectional. A user can transmit and receive information at the same time.

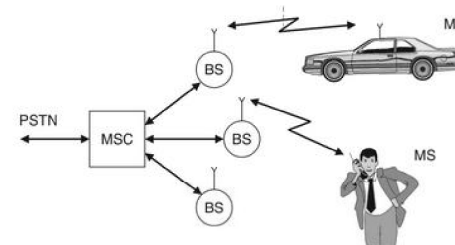


Figure Principle of a cellular system. In this figure: PSTN, Public Switched Telephone Network; MSC, Mobile Switching Center; BS, Base Station; MS, Mobile Station.

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Trunking Radio

Trunking radio systems are an important variant of cellular phones, where there is no connection between the wireless system and the public switched telephone network (PSTN); therefore, it allows the communications of closed user groups. Obvious applications include police departments, fire departments, taxis, and similar services. The closed user group allows following operations

1. Group calls
2. Call priorities
3. Relay networks

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Wireless Local Area Networks

- The functionality of Wireless Local Area Networks (WLANs) is very similar to that of cordless phones connecting a single mobile user device to a public landline system.
- A major difference between wireless LANs and cordless phones is the required data rate. While cordless phones need to transmit (digitized) speech, which requires at most 64 Kbit/s, wireless LANs should be at least as fast as the Internet that they are connected.

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Cordless phone

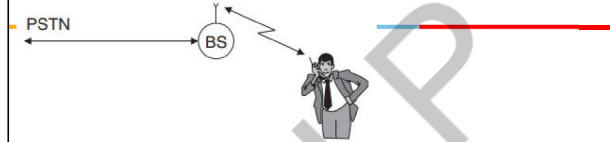


Figure Principle of a simple cordless phone.

Wireless PABX

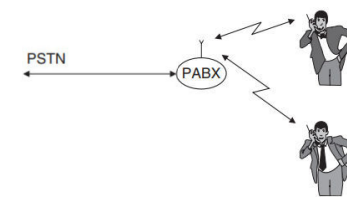


Figure Principle of a wireless private automatic branch exchange.

Personal Area Networks

- When the coverage area becomes even smaller than that of WLANs, we speak of Personal Area Networks (PANs).
- Such networks are mostly intended for simple cable replacement duties.

For example, devices following the Bluetooth standard allow connecting a hands-free headset to a phone without requiring a cable; in that case, the distance between the two devices is less than a meter.

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Fixed Wireless Access

- Fixed wireless access systems can also be considered as a derivative of cordless phones or WLANs, essentially replacing a dedicated cable connection between the user and the public landline system.

The main difference from a cordless system is that

- there is no mobility of the user devices and
- the BS almost always serves multiple users.

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Requirements for the Services

- Data Rate
- Range and Number of Users
- Mobility
- Energy Consumption
- Use of Spectrum

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Data Rate

Data rates for wireless services ranges from few bits per second to several gigabits per second, depending on the application.

Type of application	Data rate
Sensor networks	1 Kbit/s
Speech communications	5 and 64 Kbit/s
Elementary data services	10 and 100 Kbit/s
Computer peripherals and similar devices	1Mbit/s
High-speed data services	0.5 to 100 Mbit/s
Personal Area Networks	100 Mbit/s

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Range and Number of Users

Another distinction among the different networks is the range and the number of users that they serve. By range, we mean here the distance between transmitter and receiver.

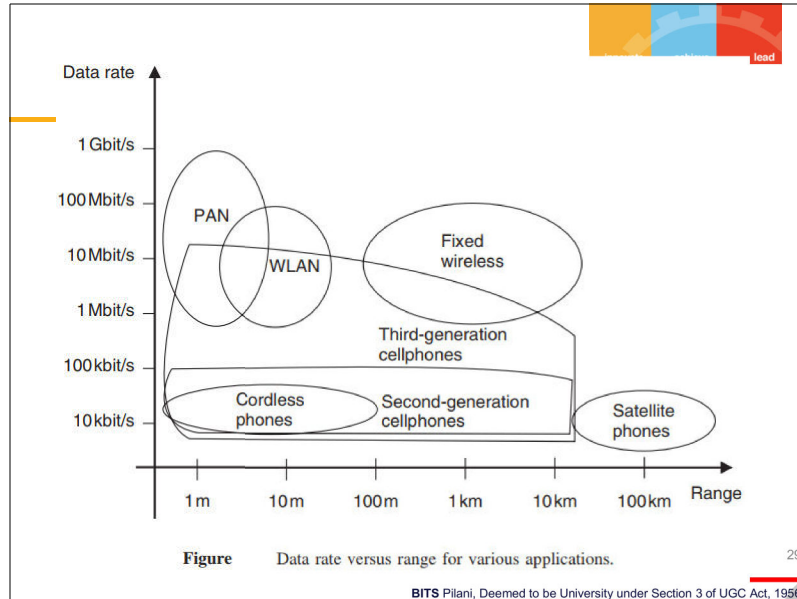
The coverage area of a system can be made almost independent of the range by just combining a larger number of BSs into one big network.

Type of network	Range
Body Area Networks (BANs)	1meter
Personal Area Networks	10 meter
WLANs	up to 100 meter
Cellular systems	10 or even 30 km
Satellite systems	1,000 km radius
Fixed wireless access services	100 m and several tens of kilometers

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Mobility

- Wireless systems also differ of mobility that they have to allow for the users.
- The ability to move around while communicating is one of the main factors of wireless communication for the user.

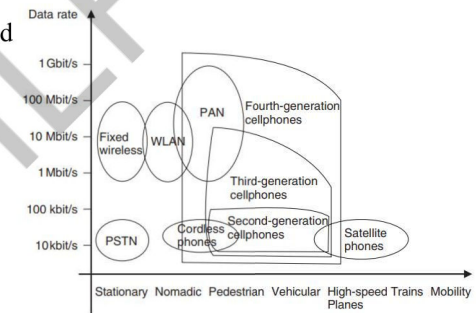


Figure 30 Data rate versus mobility for various applications.

Energy Consumption

Energy consumption is a critical aspect of wireless devices. Most wireless devices use (one way or rechargeable) batteries, as they should be free of any wires both the ones used for communication and the ones providing the power supply.

- Rechargeable batteries
- One way batteries
- Power mains

Use of Spectrum

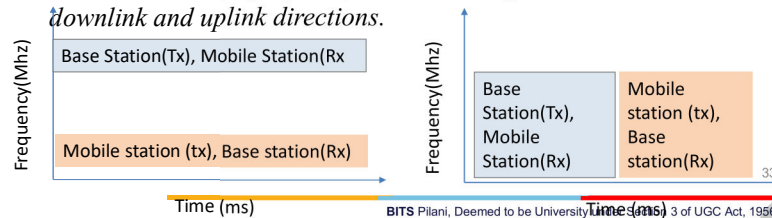
Spectrum can be assigned on an exclusive basis, or on a shared basis. That determines to a large degree the multiple access scheme and the interference resistance that the system has to provide:

- Spectrum dedicated to service and operator
- Spectrum allowing multiple operators
- Spectrum dedicated to a service
- free spectrum
- Adaptive spectral usage



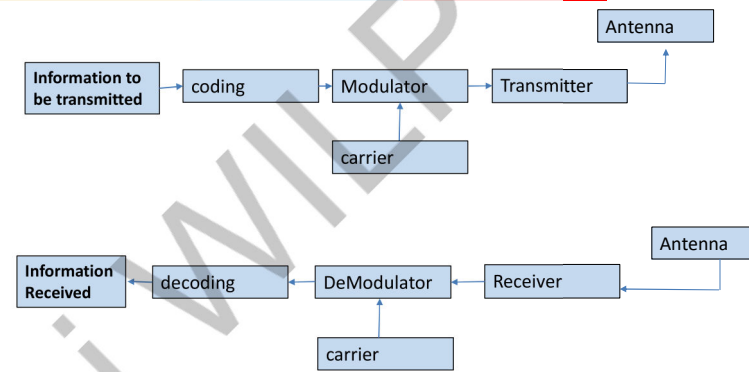
Direction of Transmission (The duplex concept)

- *Simplex systems*— e.g., broadcast systems and pagers.
- *Semi-duplex systems* – e.g., Walkie-talkies
- *Full-duplex systems* – e.g., cellphones and cordless phones.
- **FDD** uses two separate frequencies for the uplink (from the mobile to the BS) and the downlink (from the BS to the mobile).
- **TDD** uses a single frequency to transmit signals in both the downlink and uplink directions.



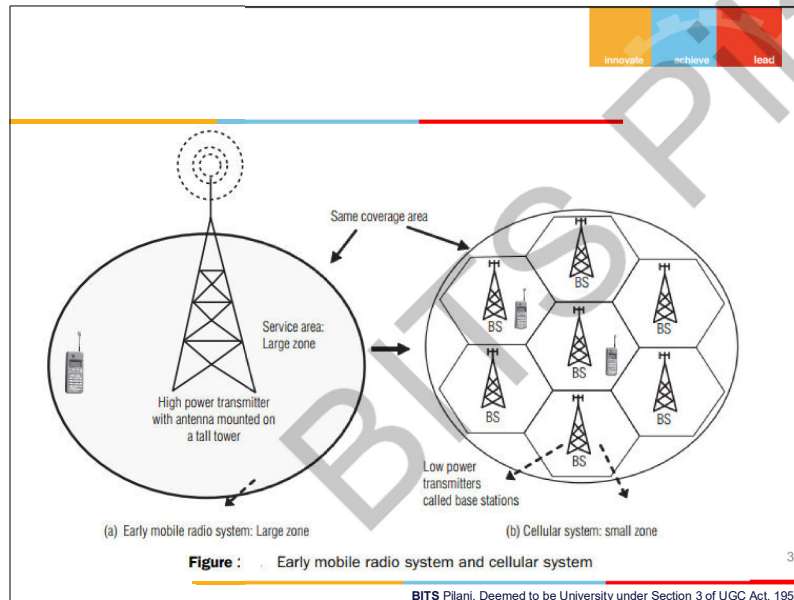
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Wireless communication systems

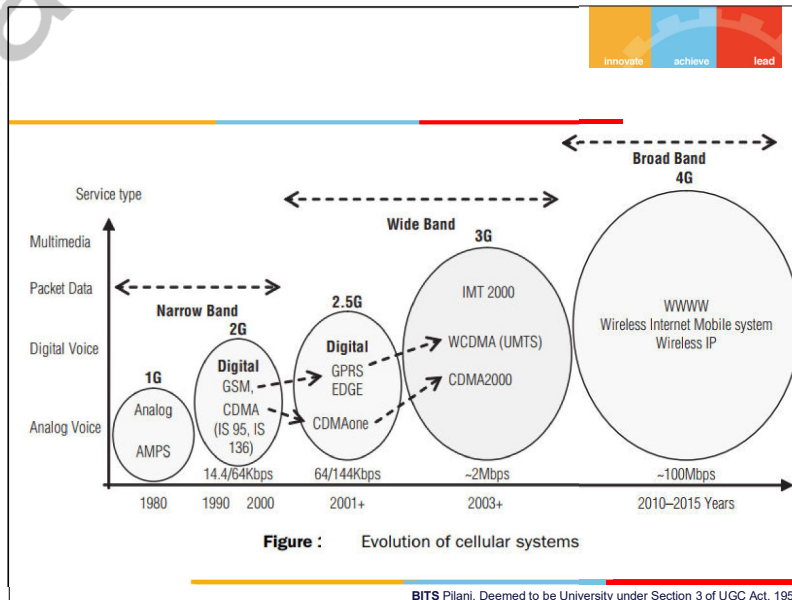


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Generations	1G	2G	3G	4G	5G
Deployment	1979	1991	2001	2009	2018
Data rate	2.4Kbps	64Kbps	2Mbps	100Mbps	10Gbps
	Basic Voice Service	Designed for voice	Voice and Data	IP-based protocols	Low latency
	Analog based	Digital standards	Mobile broadband	True mobile broadband	Fast speed
Switching Techniques	Circuit switching	Circuit switching	Circuit and packet switching	packet switching	packet switching
Access Techniques	FDMA	GSM, TDMA, CDMA	WCDMA, UMTS, CDMA 2000, HSPA/HSDPA	LTEA, OFDMA, SC-FDMA, WiMAX	BDMA, NOMA FBMC
Application	Voice	Voice and data	Voice, data, and video calling	Voice, data, video calling, HD television, etc.	Voice, data, video calling, ultra HD video, VR application
Core network	PSTN	PSTN & packet	Internet	Internet	Internet
Data rate	2.4 kbps	14.4 kbps	384 kbps to 2 Mbps	100 Mbps to 500 Mbps	10 Gbps to 50 Gbps
Frequency band	800 MHz	800 MHz, 900 MHz, 1800 MHz, 1900 MHz	800 MHz, 900 MHz, 1800 MHz, 1900 MHz, 2100 MHz	2–8 GHz	1.8 GHz, 2.6 GHz, 30–300 GHz
Advantages	Mobility	Digital voice calling, mass adoption, low power consumption	Better internet experience, support of both voice and video calling	High data rate, body area network	Low latency, higher data rate, wider coverage
Disadvantages	Poor security, low capacity, low coverage	Low data rate, inability to support video transmission	Expensive terminals and high-power consumption	Expensive and high-power consumption	Requires many small cell antennas
Main Services Provided	Voice Calls	SMS and Voice	Web applications, Video Services, TV streaming	Mobile application, HDTV, DVB	IoT, Smart City, etc., Vulnerabilities in SDN, NFV, and cloud techniques, the openness of the network
Security Issues	No guarantee of security and privacy	Authentication is one-way, illegal devices	The encryption Key, Vulnerabilities in IP traffic	Vulnerabilities in the MAC layer, threats from new devices	

Cellular Network- 1G

Goal: To develop a working system that could provide basic voice service

Time frame: 1980s in US

Technology: FDMA/FDD

Speed: 2.4Kbps

1G network used Analogue circuit-switched technology with FDMA 800–900 MHz frequency bands.

Example Systems:

- Analogue Mobile Phone System (AMPS-USA)
- Total Access Communication System (TACS-UK)
- Nordic Mobile Telephone (NMT-Europe)

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In AMPS, two 25-MHz bands are allocated. One 25-MHz band is for communication from BS to mobile unit and the other for communication from mobile unit to BS.

The following are the limitations of 1G:

- Supports only speech
- Low traffic capacity
- Unreliable handover
- Long-call setup time and frequent call drops
- Inefficient use of bandwidth and poor battery life
- Poor voice quality and large phone size
- Allows users to make voice calls in 1 country only



- the first handheld device from Motorola Company which was available in 1984 in 1G network.

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Cellular Network-2G

- Goal: Digital voice service with improved quality and also provide better services
- Time Frame: Launched in Finland in 1991
- 2G technology supports data, speech, FAX, SMS, and WAP services.
- The frequency bands used by GSM are 890–960 MHz and 1710–1880 MHz. (890–915 MHz - uplink MS to BS, 935–960 MHz downlink BS to MS)
- Used digital signals

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Digital modulation formats were introduced in this generation with the main technology as TDMA/FDD

Data Speed: upto 64kbps

Services like text message, picture message was included.

Example Systems:

- Global System for Mobile Communications (GSM) (TDMA and FDMA)
- IS-95(Interim Standard) (CDMA)

Ericsson GH218 which was introduced in 1994 and operated in 2G networks.



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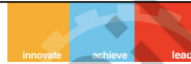


The following are limitations of 2G:

- Provides low data rates ranging from 9.6 kbps to 28.8 kbps.
- Circuit-switched network, where the end systems are dedicated for the entire call session. This causes reduction in usage of bandwidth and resources.
- Too many 2G standards globally (e.g. GSM, CDMA, PDC, and PHS)

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Interim generation (2.5G)



- Goal: To provide better data rates and wider range of data services and it was a transition from 2G to 3G
- Time frame: 2000-2002
- *The mobile technology using GPRS standard has been termed as 2.5G.*
- The 2.5G was started in 1998 with added GPRS and enhanced data rates for GSM evolution (EDGE).

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Cellular Network- 3G



- The aim of 3G systems is to provide communication services from person-to-person at any place (global roaming) and at any time through any medium with guaranteed quality of service.
- high-speed internet access, live video communications, and simultaneous data and voice transmission .
- The 3G technology has added multimedia facilities to 2.5G phones.
- 3G operates in the frequency band of 1710–2170 MHz.
- It provides high transmission rates from 384 Kbps in a moving vehicle to 2 Mbps for stationary or mobile users.

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- Examples of 3G system are universal mobile telecommunication systems (UMTS) and international mobile telecommunications at 2,000 MHz (IMT-2000).
- UMTS are designed to provide different types of data rates, based on the following circumstances:
 - up to 144 kbps for moving vehicles
 - 384 kbps for pedestrians
 - 2 Mbps for indoor or stationary users



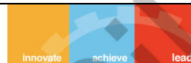
LG U8110 that was introduced in 2004 and is operating in 3G networks.



The following are the drawbacks of 3G system:

- High bandwidth requirement
- High spectrum licensing fees
- Expense and bulk size of 3G phones
- Lack of 2G mobile user buy in for 3G wireless service
- Lack of network coverage because it is still a new service
- High prices of 3G mobile services in some countries

Cellular Network- 4G



- Goal: High mobility, High data rate
- Frequency Band: 2 to 8 Ghz.
- Download speed is upto 100 Mbps for moving users and 1Gbps for stationary users.

The following modulation techniques are proposed to be used in the 4G cellular phones.

Variable spreading factor-orthogonal frequency and code division multiplexing (VSF-OFCDM).

Variable spreading factor code-division multiple access (VSF- CDMA).

LTE (Long Term Evolution)

- ✓ Uses Orthogonal Frequency Division Multiplexing (OFDM)
- ✓ Uses Multi-input Multi-output(MIMO) for enhanced throughput

Features of 4G



- Support for interactive multimedia, voice, streaming video, Internet, and other broadband services
- Fully IP based mobile system
- High speed, high capacity, and low cost-per-bit
- Global access, service portability, and scalable mobile services
- Seamless switching, and a variety of Quality of Service
- Better scheduling and call-admission-control techniques
- Better spectral efficiency
- Interoperability with existing wireless standards
- Packet switched network



Advantages of 4G

- Affordable communication services.
- High speed, high capacity and low cost per bit.
- Support for interactive multimedia, voice, streaming video, Internet and other broadband services.
- Global access, Service portability and scalable mobile services and variety of quality of services provided.
- Better spectral efficiency.
- seamless network of multiple protocol and air interfaces.

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Table History of 1G, 2G, 3G, and 4G technologies

Technology	Various generations				
	1G	2G	2.5G	3G	4G
Design began	1970	1980	1985	1990	2000
Implementation	1984	1991	1999	2002	2012–2015
Service	Analogue voice	Digital voice	High-capacity packets, MMS	High-capacity broadband data	Higher capacity, completely IP, Multimedia
Multiple access	FDMA	TDMA, CDMA	TDMA, CDMA	CDMA	OFDMA
Standards	AMPS, TACS, NMT	CDMA, GSM, PDC	GPRS, EDGE	WCDMA, CDMA2000	Single standard
Bandwidth	1.9 kbps	14.4 kbps	384 kbps	2 Mbps	200 Mbps
Core network	PSTN	PSTN	PSTN, Packet network	Packet network	Internet

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Handoff: A crucial component of the cellular concept is the notion of handoffs. Handoff is the process of transferring an active call from one cell to another as the mobile unit moves from one cell to the other.

Handoff operation involves identifying a new BS and allocation of voice and control signals associated.

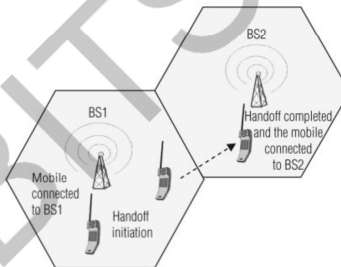


Figure Call handoff process

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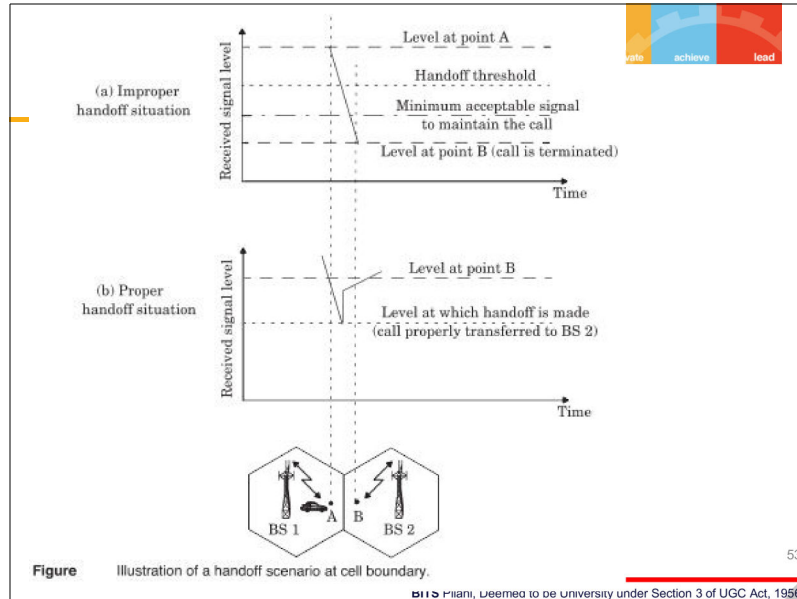
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Handoff –

- high priority over call initiation requests
- must be performed successfully
- Infrequently
- be imperceptible.
- The margin $\Delta = P_{r \text{ handoff}} - P_{r \text{ minimum usable}}$ cannot be too large or too small.

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- In deciding when to handoff, it is important to ensure that the drop in the measured signal level is not due to momentary fading.
- Information about the vehicle speed is useful in handoff decisions.
- Dwell time –The time over which a call may be maintained within a cell without handoff.
- The statistics of dwell time are important in the practical design of handoff algorithm.

- In first generation analog cellular system, the locator receiver is used for handoff.
- In second generation system, handoff decisions are mobile assisted –mobile assisted handoff (MAHO).
- Intersystem handoff –different cellular systems –roamer.

Prioritizing Handoffs

- Guard channel concept –can be used with dynamic channel assignment strategies to increase spectrum utilization.
- Queuing of handoff –possible due to the fact that there is a finite time interval between handoff threshold time and call-terminated time



Practical Handoff Considerations



- Several handoff schemes have been devised to handle the simultaneous traffic of high speed and low speed users – minimizing the handoff intervention from the MSC

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HOME WORK



- If 0 dBm is equal to 1 mW (10^{-3} W) over a 50Ω load; express 10W in units of dBm.
- In simulcasting paging systems, there usually is one dominant signal arriving at the paging receiver. In most, but not all cases, the dominant signal arrives from the transmitter closest to the paging receiver. Explain how the FM capture effect could help reception of the paging receiver. Could the FM capture effect help cellular radio systems? Explain how
- Assume a 1 Amp-hour battery is used on a cellular telephone (often called a cellular subscriber unit). Also assume that the cellular telephone draws 35 mA in idle mode and 250 mA during a call. How long would the phone work (i.e., what is the battery life) if the user leaves the phone on continually and has one 3-minute call every day? Every 6 hours? Every hour? What is the maximum talk time available on the cellular phone in this example?

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- Discuss the similarities and differences between a conventional cellular radio system and a space-based (satellite) cellular radio system. What are the advantages and disadvantages of each system? Which system could support a larger number of users for a given frequency allocation? Why? How would this impact the cost of service for each subscriber?

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- In your place of work, how many modern wireless communications networks are available to you? Identify the types of services, the types of technologies, the commercial names of the service providers, and the commercial names of the equipment manufacturers that offer these wireless access capabilities.
- In your home, how many modern wireless communications networks are available to you? Identify the types of services, the types of technologies, the commercial names of the service providers, and the commercial names of the equipment manufacturers that offer these wireless access capabilities.
- How would multicarrier transmissions impact an operator's approach to allocating resources to accommodate a growing subscriber database that increasingly desires data connectivity over voice? How would heavy HSCSD usage impact a cellular carrier's strategy in allocating channels in the base stations of a cellular network? How would the rapid adoption of Voice over Internet Protocol (VoIP) impact cellular congestion? Explain.

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IMP Note to Self



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(CSI ZG520)**

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Modular Content Structure

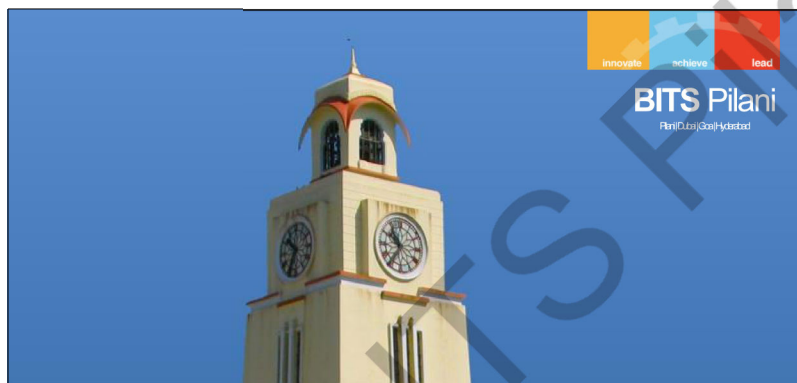
- M1: Introduction to Wireless Systems and Modern Wireless Communication System.
- M2: The Cellular Concept- System Design Fundamentals**
- M3: Cellular Network
- M4: Mobile Radio Propagation: Large scale path Loss_ Part 1
- M5: Mobile Radio Propagation: Large scale path Loss_ Part 2
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Modular Content Structure

- M9: Modulation Techniques for Mobile Radio_ Amplitude Modulation
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M2: The Cellular Concept- System Design Fundamentals

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Text Book:

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- R1: “Mobile Communication” by Jochen H. Schiller, Addison –Wesley, Pearson Education Ltd., 2000.
- R2: “Wireless and Mobile Network Architectures” by Yi-Bing Lin and Imrich Chaltamac, John Wiley and Sons, 2001.

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M1: Introduction to Wireless Systems and Modern Wireless Communication System.

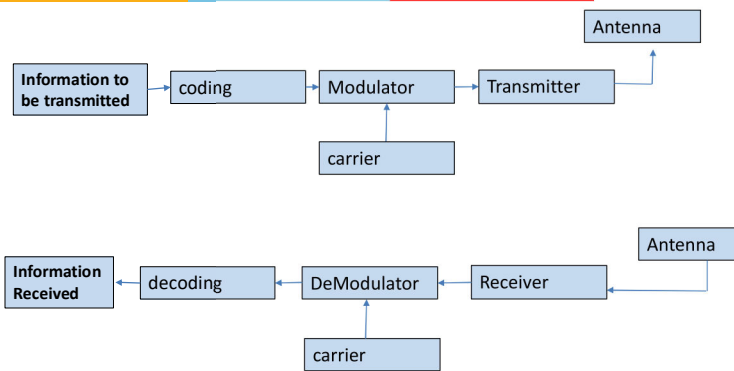
- | Type | Description/Plan/Reference |
|----------|---|
| RL 2.1.1 | Cellular Concept |
| RL 2.1.2 | Cellular concept frequency reuse |
| RL 2.1.3 | Channel Assignment strategies and Introduction to Handoff |
| RL 2.1.4 | Handoff Strategies |
| CS 2 | A brief summary of recorded lectures, Frequency reuse, Channel assignment strategies, Prioritizing handoff, Problems on frequency re-use. |
| HW 1 | T1: Problems: 3.1, 3.2, 3.3, 3.5, 3.6. |
| LE 1 | MATLAB experiment based on the frequency reuse |

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Wireless communication systems

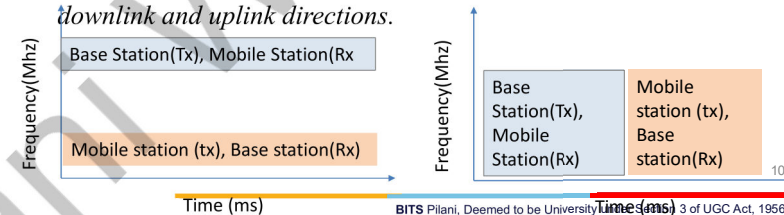


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Direction of Transmission (The duplex concept)

- *Simplex systems* – e.g., broadcast systems and pagers.
- *Semi-duplex systems* – e.g., Walkie-talkies
- *Full-duplex systems* – e.g., cellphones and cordless phones.
- **FDD** uses two separate frequencies for the uplink (from the mobile to the BS) and the downlink (from the BS to the mobile).

TDD uses a single frequency to transmit signals in both the downlink and uplink directions.



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Cellular Geometry

- The main reason for defining cells in a cellular land mobile radio system is to outline areas in which specific channels and specific cell sites are used.
- Designers have realized that visualizing all cells as having the same geometrical shape helps to ease the design of cellular systems.
- cellular geometry helps to ease the assessment of spectral efficiency of various cellular systems, in particular to calculate the significant co-channel interference (CCI) in the system.

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Cell Shapes

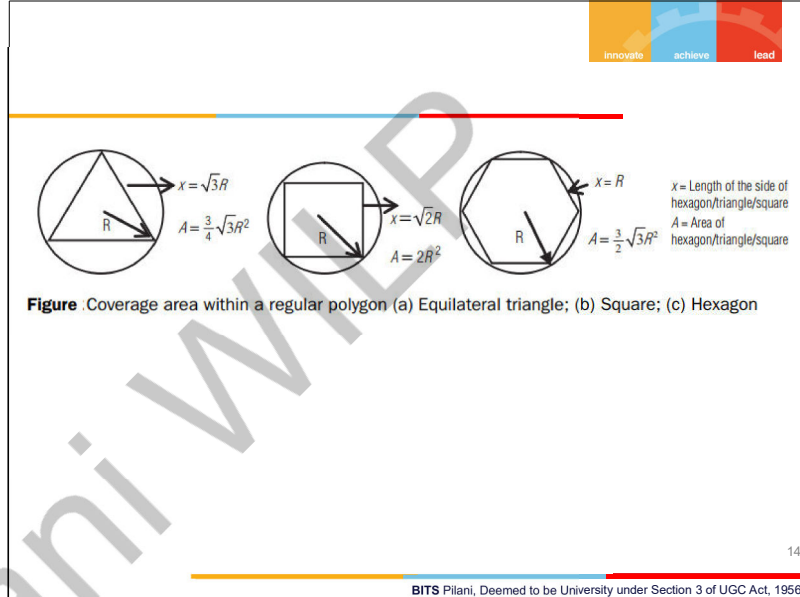
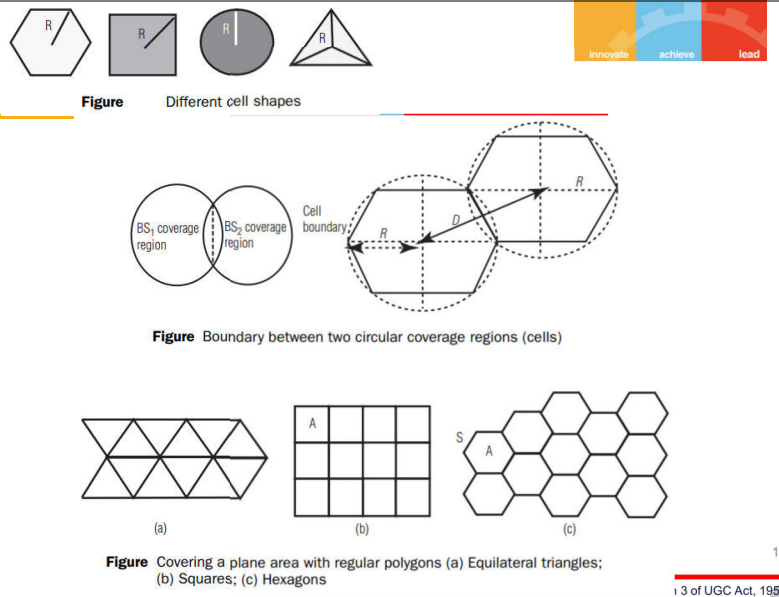
Shapes: Regular Hexagon, Square, Circle & Triangle.

The regular hexagon is favoured by system designers for the following reasons:

- It provides the best approximation to the circular omnidirectional radio patterns achieved in practice.
- It is more economical to use since a hexagonal layout requires fewer cells and hence fewer stations.
- It combines ease of geometry with the practical realization of overlapping circles.
- For a given distance between the centre of a polygon and its farthest perimeter points, the hexagon has the largest area, and it almost approximates a circular radiation pattern.

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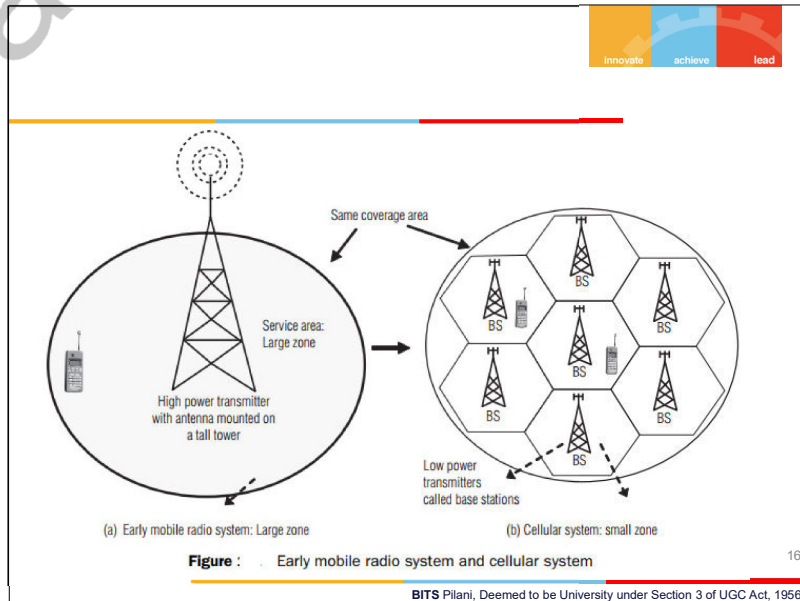


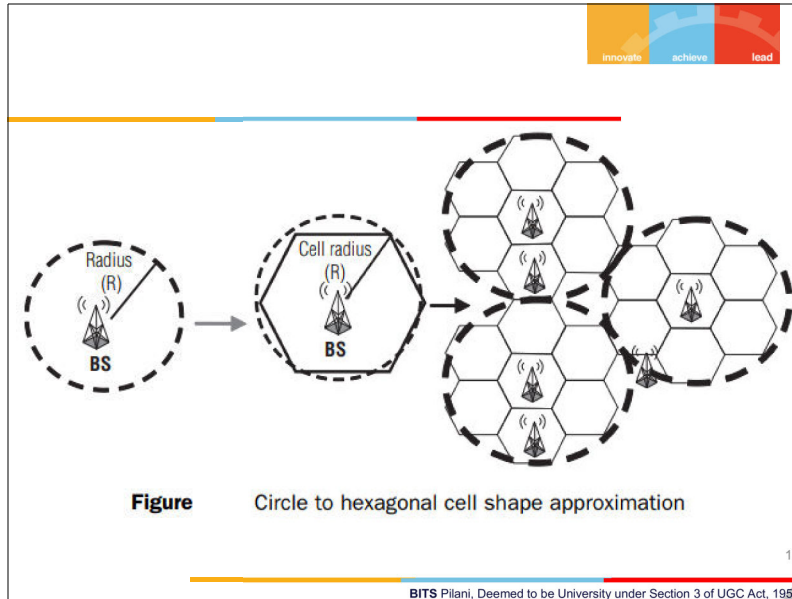
Introduction to cellular concept

The cellular concept was developed in response to the limitations of conventional mobile radio services.

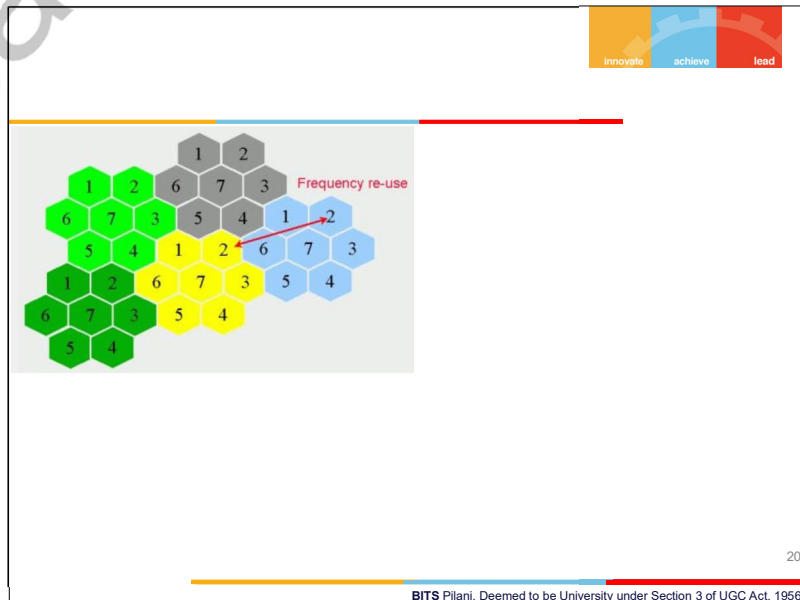
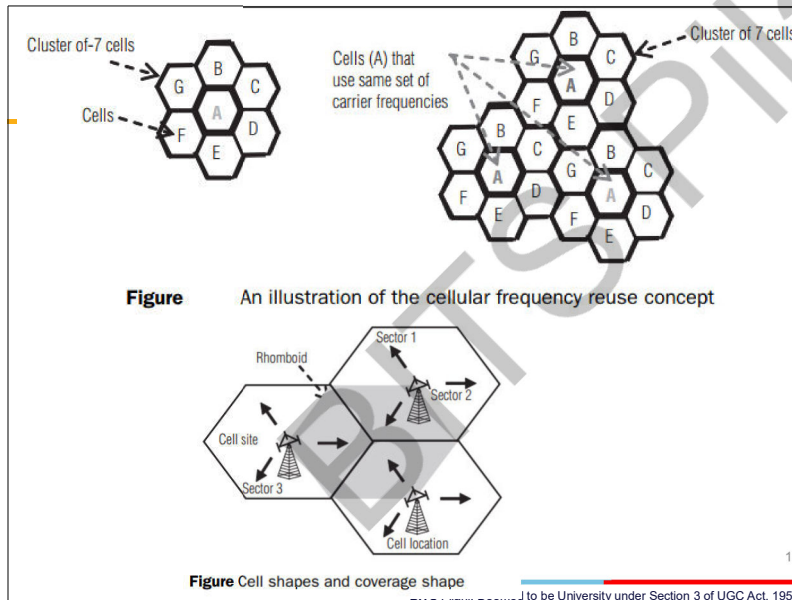
The main limitations of the previous mobile communication systems are as follows:

1. High-power transmitters were used to cover very large area.
2. Inefficient use of allocated radio spectrum.
3. If a user leaves the coverage area, they had to reinitiate the call on a different frequency channel.





Frequency reuse: This is the process in which the same set of frequencies (channels) can be allocated to more than one cell to increase the system capacity. Instead of deploying a powerful BS in a large coverage area, the area is divided into multiple smaller cells and a BS deployed in each cell can use smaller transmit power. Thus, two transmissions can employ the same frequency if they are far away enough such that the CCI level is below a desired threshold.





EX: The bandwidth allocated to a particular FDD cellular system is 33 MHz. It uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses
(a) 4 cell reuse and (b) 7 cell reuse

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Solution

Total bandwidth allocated to cellular system = 33 MHz
Channel bandwidth required for duplex channel (uplink and downlink) = $25 \text{ kHz} \times 2 = 50 \text{ kHz}$

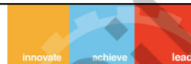
Total number of available channels = $33,000,000 / 50,000 = 660$

a) For cluster size $N = 4$, number of channels available per cell
= $660 / 4 = 165$

b) For cluster size $N = 7$, number of channels available per cell
= $660 / 7 = 95$

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Cellular system capacity and frequency reuse for a cluster size of “N” with each cell allocated a group of “K” channels



To understand the frequency reuse concept,
Consider a cellular system which has a total of S duplex channels available for use in a cluster
If each cell is allocated a set of k channels ($k < S$) and the cluster size is N
The total number of available radio channels can be expressed as

$$S = kN$$

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The N cells which collectively use the complete set of available frequencies is called a *cluster*.

If a cluster is replicated M times within the system,
The total number of duplex channels, C , can be used as a measure of capacity and is given by

$$C = MkN = MS$$

the capacity of a cellular system is directly proportional to the number of times a cluster is replicated in a fixed service area.

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- The factor of N is called the cluster size and is typically equal to 4, 7, or 12.
- For the same cell size at a given area, N decreases $\Rightarrow M$ increases $\Rightarrow C$ increases.

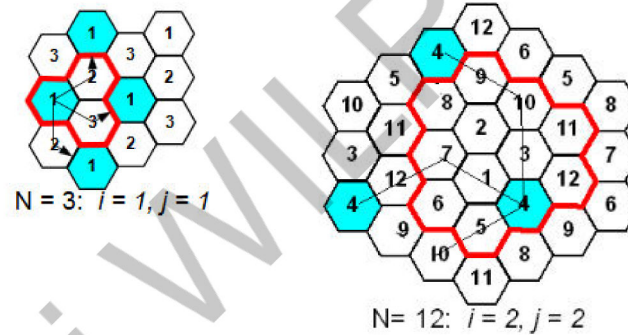
- The frequency reuse factor of a cellular system is given by $1/N$.
- N can only have values which satisfy Equation

$$N = i^2 + ij + j^2$$

where i and j are non-negative integers. To find the nearest co-channel neighbors of a particular cell, one must do the following:

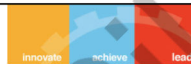
- (1) move i cells along any chain of hexagons and
- (2) Turn 60 degrees counter-clockwise and move j cells.

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Co-channel Interference



- Interference between signals from co-channels
- Increasing the signal-to-noise ratio (SNR) to overcome thermal noise does not work
- Physically separate co-channel cells
- Co-channel reuse ratio
 - $Q = \frac{D}{R} = \sqrt{3N}$
 - Condition: same cell size, same transmitter power
 - R : radius of the cell
 - D : distance between centers of the nearest co-channel cells

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Relationship between frequency reuse factor (D/R) and cluster size (N)

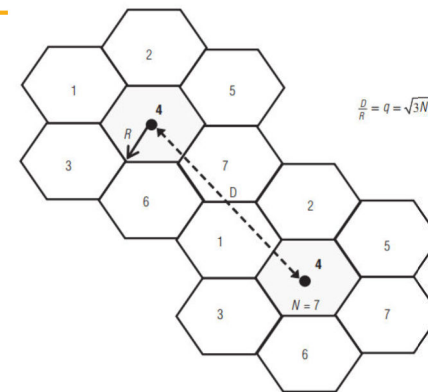


Figure Frequency reuse factor or co-channel reuse ratio (q)

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Determination of relation between frequency reuse factor (D/R ratio) and the cluster size (N) involves mainly two steps:

1. finding the relation between the distance D (two co-channel cells) and R
2. locating the co-channel cells.

First let us derive the relation between distance (D) (two co-channel cells) and R in a hexagonal geometry
The actual centre-to-centre distance between two adjacent hexagonal cells is

$$d = 2R \cos 30^\circ \text{ or } \sqrt{3}R$$

where R is the centre to vertex distance

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Proof of $d = \sqrt{3}R$

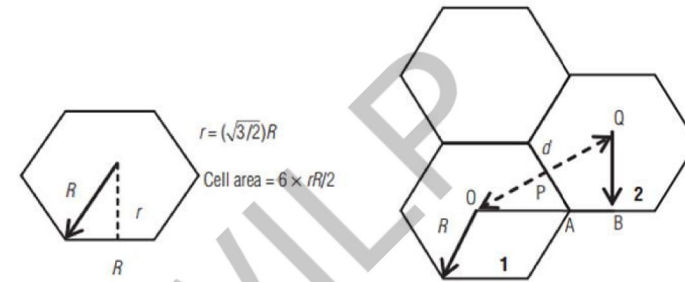


Figure Centre-to-centre distance between two adjacent hexagonal cells 1 and 2

$OA = R$, $AB = R/2$ and OAP is a right-angled triangle

In the OAP triangle, $OP = OA \sin 60^\circ = (\sqrt{3}/2)R$

$$OQ = OP + PQ \quad d = \left(\frac{\sqrt{3}}{2}\right)R + \left(\frac{\sqrt{3}}{2}\right)R = \sqrt{3}R$$

$$\text{Area of a small hexagon with radius } R = 6 \times \frac{1}{2} \times R \times \frac{\sqrt{3}}{2}R = \frac{3\sqrt{3}}{2}R^2 = 2\sqrt{3}r^2$$

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Channel Assignment Strategies

- The objectives of channel assignment strategies are to increase capacity and minimize interference.
- Channel assignment strategies are classified as either fixed and dynamic.
- In a fixed channel assignment strategy, each cell is allocated a predetermined set of voice channel.
- If all the channels in that cell are occupied, the call is blocked. A variation is the borrowing strategy.

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In a dynamic channel strategy, voice channels are not allocated to different cells permanently.

This strategy reduces the likelihood of blocking.

Mobile switching center (MSC) must collect real-time data on channel occupancy, traffic distribution, and radio signal strength indications (RSSI) of all channels.

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Handoff: A crucial component of the cellular concept is the notion of handoffs. Handoff is the process of transferring an active call from one cell to another as the mobile unit moves from one cell to the other.

Handoff operation involves identifying a new BS and allocation of voice and control signals associated.

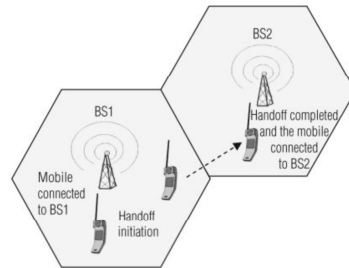


Figure Call handoff process

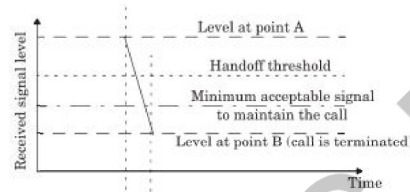
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Handoff –

- high priority over call initiation requests
- must be performed successfully
- Infrequently
- be imperceptible.
- The margin $\Delta = P_{r \text{ handoff}} - P_{r \text{ minimum usable}}$ cannot be too large or too small.

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(a) Improper handoff situation



(b) Proper handoff situation

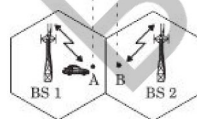
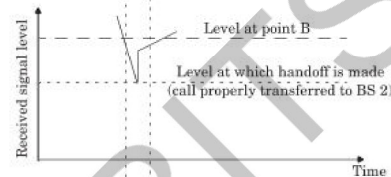


Figure Illustration of a handoff scenario at cell boundary.

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- In deciding when to handoff, it is important to ensure that the drop in the measured signal level is not due to momentary fading.
- Information about the vehicle speed is useful in handoff decisions.
- Dwell time –The time over which a call may be maintained within a cell without handoff.
- The statistics of dwell time are important in the practical design of handoff algorithm.

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- In first generation analog cellular system, the locator receiver is used for handoff.
- In second generation system, handoff decisions are mobile assisted –mobile assisted handoff (MAHO).
- Intersystem handoff –different cellular systems –roamer.

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Prioritizing Handoffs

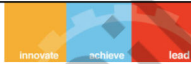


- Guard channel concept –can be used with dynamic channel assignment strategies to increase spectrum utilization.
- Queuing of handoff –possible due to the fact that there is a finite time interval between handoff threshold time and call-terminated time

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Practical Handoff Considerations



- Several handoff schemes have been devised to handle the simultaneous traffic of high speed and low speed users – minimizing the handoff intervention from the MSC

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HOME WORK



- 3.1 Prove that for a hexagonal geometry, the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + ij + j^2$. Hint: Use the cosine law and the hexagonal cell geometry.
- 3.2 If two independent voltage signals, $v_1(t)$ and $v_2(t)$, are added together to provide a new resulting signal, prove that under certain conditions the resulting signal has the same power as the sum of the individual powers. What are these conditions? What special conditions must apply for this result to be valid when the signals are uncorrelated?

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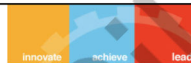




3.3 Show that the frequency reuse factor for a cellular system is given by k/S , where k is the average number of channels per cell and S is the total number of channels available to the cellular service provider.

3.5 A cellular service provider decides to use a digital TDMA scheme which can tolerate a signal-to-interference ratio of 15 dB in the worst case. Find the optimal value of N for (a) omnidirectional antennas, (b) 120° sectoring, and (c) 60° sectoring. Should sectoring be used? If so, which case (60° or 120°) should be used? (Assume a path loss exponent of $n=4$ and consider trunking efficiency.)

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- (c) Determine the forward link C/I for the following frequency reuse designs: $N=1$, $N=3$, $N=4$, $N=7$, and $N=12$. Assume that the propagation path loss exponent is four, and evaluate the S/I contribution due to just the first tier and then due to additional outer tiers of co-channel cells. Indicate the number of tiers at which there is a diminishing contribution to the interference at the mobile.
- (d) Repeat part (c), except now consider a line-of-sight path loss exponent of $n=2$. Notice the huge impact that the propagation path loss exponent has on C/I. What can you say about the cluster size, path loss exponent, and the C/I values which result? How would this impact practical wireless system design?

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3.6 You are asked to determine the signal-to-interference ratio (SIR or C/I) on the forward link of a cellular system, when the mobile is located on the fringe of its serving cell. Assume that all cells have equal radii, and that base stations have equal power and are located in the centers of each cell. Also assume that each cell transmits an independent signal, such that interfering signal powers may be added. Let us define a "tier" of cells as being the collection of co-channel cells that are more-or-less the same distance away from the mobile in the serving cell. This problem explores the impact of the cluster size (i.e., frequency reuse distance), the number of tiers used in the calculation of C/I and the effect of the propagation path loss exponent on C/I.

- (a) What is the average distance (in terms of R) between the mobile on the fringe of the serving cell and the first tier of co-channel cells? (These cells are called the "nearest neighbors.") How many cells are located in the first tier? Solve for the case of $N=1$, $N=3$, $N=4$, $N=7$, and $N=12$ cluster sizes. How does the average distance compare to the value of $D=QR$, where $Q=\sqrt{3N}$?
- (b) What is the average distance (in terms of R) between the mobile on the fringe of the serving cell and the second and third tier of co-channel cells, and how many cells are in the second and third tier of co-channel cells for the cases of $N=1$, $N=3$, $N=4$, $N=7$, and $N=12$ cluster sizes?

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IMP Note to Self





IMP Note to Self



IMP Note to Students

- It is important to know that just login to the session does not guarantee the attendance.
- Once you join the session, continue till the end to consider you as present in the class.
- IMPORTANTLY, you need to make the class more interactive by responding to Professors queries in the session.
- **Whenever Professor calls your number / name ,you need to respond, otherwise it will be considered as ABSENT**



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M3: Cellular Network

M3: Cellular Network

Type Description/Plan/Reference

RL 2.2.1 Interference and System Capacity

RL 2.2.2 Trunking and Grade of Service

RL 2.2.3 Improving the coverage and capacity in cellular system

CS 3 A brief summary of recorded lectures, channel capacity, Trunking and grade of service, Interference and system capacity.

HW 1 T1: Problems: 3.8, 3.10, 3.12.

LE 1 MATLAB experiment based on the CS 3.0

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